

Gentrification Risk Forecasting Project – Summer 2025

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As a rising senior at Purdue University majoring in Actuarial Science and Statistics, I originally had the goal of landing an actuarial internship for this summer. Despite passing both the P and FM exams and actively applying, I came up short. At the end of my spring semester, I began working as an Inside Sales Agent for the Big Red Team as well as getting ahead and studying for FAM-S (after receiving UEC credit for FAM-L in the spring). After my final exams I realized I would have the time to take on a project that would not only reflect my technical training, but also push me beyond the classroom and give myself some real-world experience with messy data, despite not receiving an internship opportunity.

I've always had a strong interest in geospatial analysis, though I hadn't formally studied it. This project became my way to explore that space while drawing on the core skills I've built in statistics, time series forecasting, Excel/VBA, and data validation, all with the help of AI and learning as I went. I wanted to create something real and somewhat novel, so I built a predictive model and interactive map that forecasts gentrification risk by census tract in Indianapolis (Marion County, Indiana), the closest major city to me.

Project Goals

The central goal was to go beyond static descriptions of gentrification and instead forecast where it is most likely to intensify in the coming years. This meant designing a tool that had these three characteristics:

- **Data-driven** and transparent in logic,
- **Forward-looking**, not just descriptive,
- **Accessible**, capable of serving researchers, investors, and policy stakeholders.

Importantly, I aimed for the project to acknowledge both sides of the gentrification conversation. I recognized the potential for reinvestment while also grappling with the very real displacement pressures it places on historically rooted communities. Gentrification is typically a complex and painful process, particularly when rising costs make it difficult for long-time residents to remain in their neighborhoods. While this tool leans more toward identifying economic trends and investment opportunities (or even informing urban planning decisions), it's critical to acknowledge the social consequences and equity concerns that commonly comes with that growth.

Workflow & Process

I began by researching academic literature and articles to identify the most predictive indicators of gentrification. I ranked them by both apparent significance and feasibility to obtain, knowing I had to work with publicly available data. The 8 selected were what I had determined best fit the criteria.

Using the tidycensus package in R, I pulled data for all census tracts in Marion County from 2015 to 2023. I then acquired Zillow and MAC Calls data and used a crosswalk to get them mapped onto census tracts. After that I applied ARIMA forecasting through 2028 using a custom robust function I developed to handle edge cases and fallback safely ensure stability. I also implemented LOESS smoothing, capped outlier z-scores, and created data integrity filters. Only clean, sufficient time series were used. I then normalized forecast values and year-over-year deltas into z-scores, enabling consistent comparisons across metrics. I then made sure all raw data had 2023 values for my 2023 baseline and used interpolation if missing. This was followed by a shift into Excel, where I built automated VBA scripts to flag and zero out invalid data and bad data. Then finally one last script to compile composite risk scores per tract and year.

My scoring formula:

Final score = $0.5 \times \text{Delta Mean} + 0.25 \times \text{Forecast Mean} - 0.25 \times \text{2023 Baseline Mean}$

This structure places highest weight on recent growth. It also rewards improving low-baseline tracts for having room for growth and penalizes stagnation in already-gentrified areas.

Mapping & Visualization

To communicate my findings, I developed an interactive web map using Mapbox GL JS with a few user-friendly features. I built everything from scratch in VS Code using HTML, CSS, and JS, managing data and version control with GitHub. This was my first full-stack mapping app and I tried to make it user friendly.

How I Used AI

Throughout, I used ChatGPT as a learning and refinement tool. I did not use it to generate the core ideas or workflows, but to help translate them into working scripts. It assisted with package selection, debugging, syntax decisions, and edge-case logic. The technical direction, modeling decisions, and design through all parts of the project came from me. AI accelerated my process and helped raise the quality of the code I wrote.

Challenges, Caveats, and What's Next

This is still clearly an early-stage tool. Forecasts are only as reliable as the data and simple ARIMA process feeding them, and tract-level resolution comes with tradeoffs. ZORI, ZHVI and 311 calls obviously are not available data in terms of census tracts, so the crosswalks are not perfect. Also, some metrics such as 311 calls are only proxies for gentrification, there would be infinite factors and context necessary to consistently predict gentrification, so this project is in essence a rough gauge. Web development was brand new to me, and I learned on the go. I have some experience in Excel, Visual Basic and R studio but generally just in classwork, most of the packages and processes were brand new. There may be minor errors and omissions throughout, but broadly it is logical and double checked.

That said, I'm proud of the outcome. It's a fully functional, technically grounded, and socially relevant project that applies to what I've learned in a real-world context. Eventually I hope to continue improving the program with:

- Automate data wrangling and program updating when new data is released.
- Use more indicators, further advanced forecasting as well as more advanced composition of a final score.

More than anything, this project is a starting point. It reflects where I am now and where I want to grow, whether that's in graduate study or a technical role relating to actuarial science, risk modeling or any interesting opportunity that presents itself. I'm excited to keep learning and to build things that connect quantitative analysis with human impact.